

Parseval's Identity

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Fourier Series

Previously on...

If f and g are periodic, period $2L$, then

$$f(x) = \sum_{n \in \mathbb{Z}} a_n e^{i \frac{n\pi x}{L}}, \quad a_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i \frac{n\pi x}{L}} dx$$

$$g(x) = \sum_{n \in \mathbb{Z}} b_n e^{i \frac{n\pi x}{L}}, \quad b_n = \frac{1}{2L} \int_{-L}^L g(x) e^{-i \frac{n\pi x}{L}} dx$$

"think": the set $\{e^{i \frac{n\pi x}{L}} \mid n \in \mathbb{Z}\}$ is a "basis" for the vector space of functions with period $2L$.

a real basis requires finite sums

the idea

$$f \mapsto \begin{pmatrix} \vdots \\ a_2 \\ a_1 \\ a_0 \\ a_{-1} \\ a_{-2} \\ \vdots \end{pmatrix} \begin{matrix} \nwarrow e^{i \frac{2\pi x}{L}} \\ \nwarrow e^{i \frac{\pi x}{L}} \end{matrix}$$

$$g \mapsto \begin{pmatrix} \vdots \\ b_2 \\ b_1 \\ b_0 \\ b_{-1} \\ b_{-2} \\ \vdots \end{pmatrix}$$

$$\begin{aligned} v, w \in \mathbb{C}^n \\ \langle v, w \rangle &= \sum_{i=1}^n v_i \overline{w_i} \\ &= v \cdot \overline{w} \end{aligned}$$

$$\langle f, g \rangle \mapsto \sum_{n \in \mathbb{Z}} a_n \overline{b_n}$$

Question: relation between this and the original f and g

Parseval's Identity

$$\sum_{n \in \mathbb{Z}} a_n \bar{b}_n = \frac{1}{2L} \int_{-L}^L f(x) \overline{g(x)} dx$$

proof: $\frac{1}{2L} \int_{-L}^L f(x) \overline{g(x)} dx = \frac{1}{2L} \sum_{m,n \in \mathbb{Z}} a_n \bar{b}_n \int_{-L}^L e^{i \frac{m\pi x}{L}} \cdot e^{-i \frac{n\pi x}{L}} dx$

$\rightarrow 2L \delta_{m,n}$

■ (m=n) $\int_{-L}^L dx = 2L$

■ (m ≠ n) $\int_{-L}^L e^{i \frac{(m-n)\pi x}{L}} dx = \left[\frac{-iL}{(m-n)\pi} e^{i \frac{(m-n)\pi x}{L}} \right]_{-L}^L$

$= A \left((e^{i\pi})^{m-n} - (e^{i\pi})^{n-m} \right) = A \left((-1)^{m-n} - (-1)^{n-m} \right) = 0$

example:

$f(x) = x^2$ on $(-\pi, \pi)$ extended periodically



$a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 e^{inx} dx = \frac{1}{2\pi} \left(-\frac{i}{n} x^2 + \frac{2}{n^2} x + \frac{2i}{n^3} \right) e^{inx} \Big|_{-\pi}^{\pi}$

+	x^2	e^{inx}
-	$2x$	$-\frac{i}{n} e^{inx}$
+	2	$-\frac{i}{n^2} e^{inx}$
	0	$\frac{i}{n^3} e^{inx}$

$= \frac{1}{\pi n^2} x e^{inx} \Big|_{-\pi}^{\pi}$

$= \frac{1}{\pi n^2} \left(\pi \underbrace{(e^{i\pi})^n}_{(-1)^n} + \pi \underbrace{(e^{i\pi})^{-n}}_{(-1)^{-n}} \right)$

$= \frac{2(-1)^n}{n^2}$

$a_0 = \frac{\pi^2}{3}$

$a_0 = \frac{\pi^2}{3}, a_n = (-1)^n \cdot \frac{2}{n^2}$

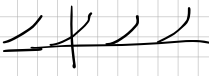
$\hookrightarrow \sum_{n \in \mathbb{Z}} a_n^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f^2(x) dx$

$\sum_{n=-\infty}^{-1} \frac{4}{n^4} + \frac{\pi^4}{9} + \sum_{n=1}^{\infty} \frac{4}{n^4} = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^4 dx = \frac{\pi^4}{5}$

~~$\sum_{n=-\infty}^{-1} \frac{4}{n^4} + \frac{\pi^4}{9} + 2 \sum_{n=1}^{\infty} \frac{4}{n^4}$~~ $= \frac{1}{2\pi} \int_{-\pi}^{\pi} x^4 dx = \frac{\pi^4}{5}$

$$\Rightarrow 8 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{4\pi^4}{45} \quad \zeta(4) = \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

example:

$f(x) = e^x$ on $(-\pi, \pi)$ extended periodically 

$$a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(1+in)x} dx = \frac{(-1)^n}{2\pi} \cdot \frac{(e^{\pi} - e^{-\pi})}{1+in}$$

$$\rightarrow |a_n|^2 = \frac{(e^{\pi} - e^{-\pi})^2}{4\pi^2} \cdot \frac{1}{1+n^2}$$

$$\frac{\cancel{(e^{\pi} - e^{-\pi})}^2}{4\pi^2 \cdot 2\pi} \sum_{n \in \mathbb{Z}} \frac{1}{n^2+1} = \cancel{\frac{1}{2\pi}} \int_{-\pi}^{\pi} e^{2x} dx = \frac{1}{2} (e^{2\pi} - e^{-2\pi})$$

$$= \frac{1}{2} (e^{\pi} - e^{-\pi})(e^{\pi} + e^{-\pi})$$

$$\sum_{n \in \mathbb{Z}} \frac{1}{n^2+1} = \frac{\pi(e^{\pi} + e^{-\pi})}{e^{\pi} - e^{-\pi}} = \pi \coth(\pi)$$

further ideas:

$$f(x) = xe^x$$

$$f(x) = \sin(x)$$